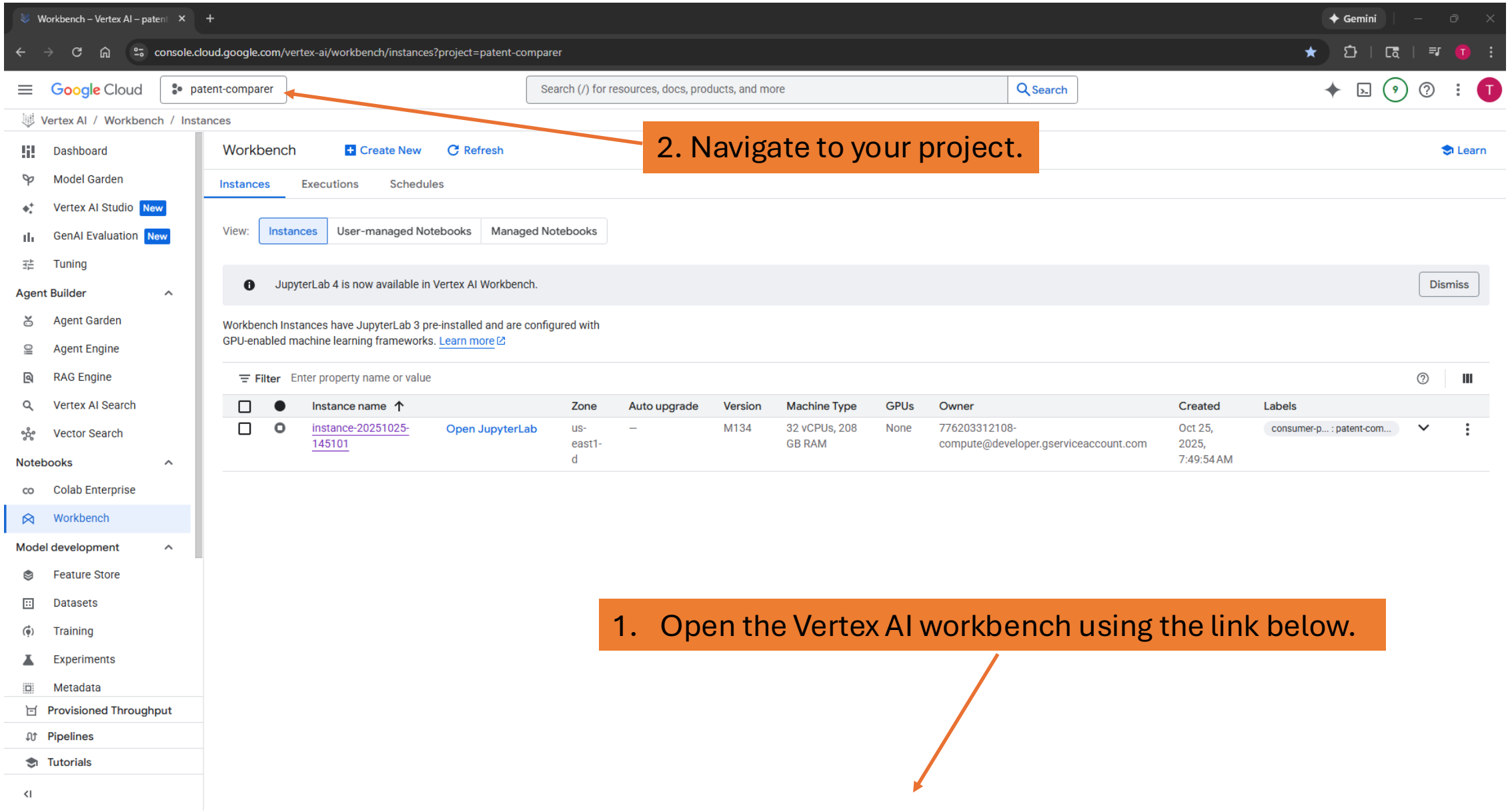
AI Patent Search Tool
Quick Start Guide

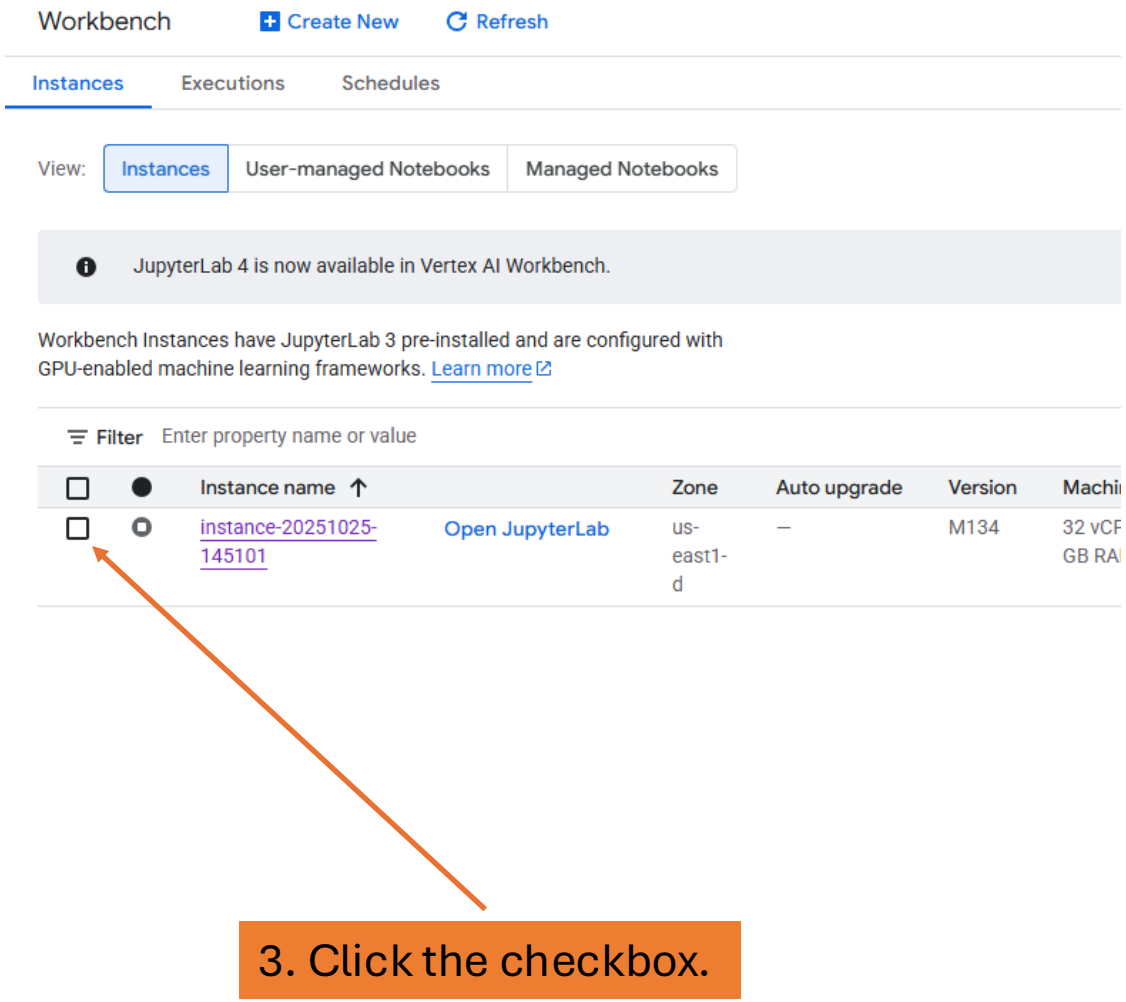
PURDUE UNIVERSITY®

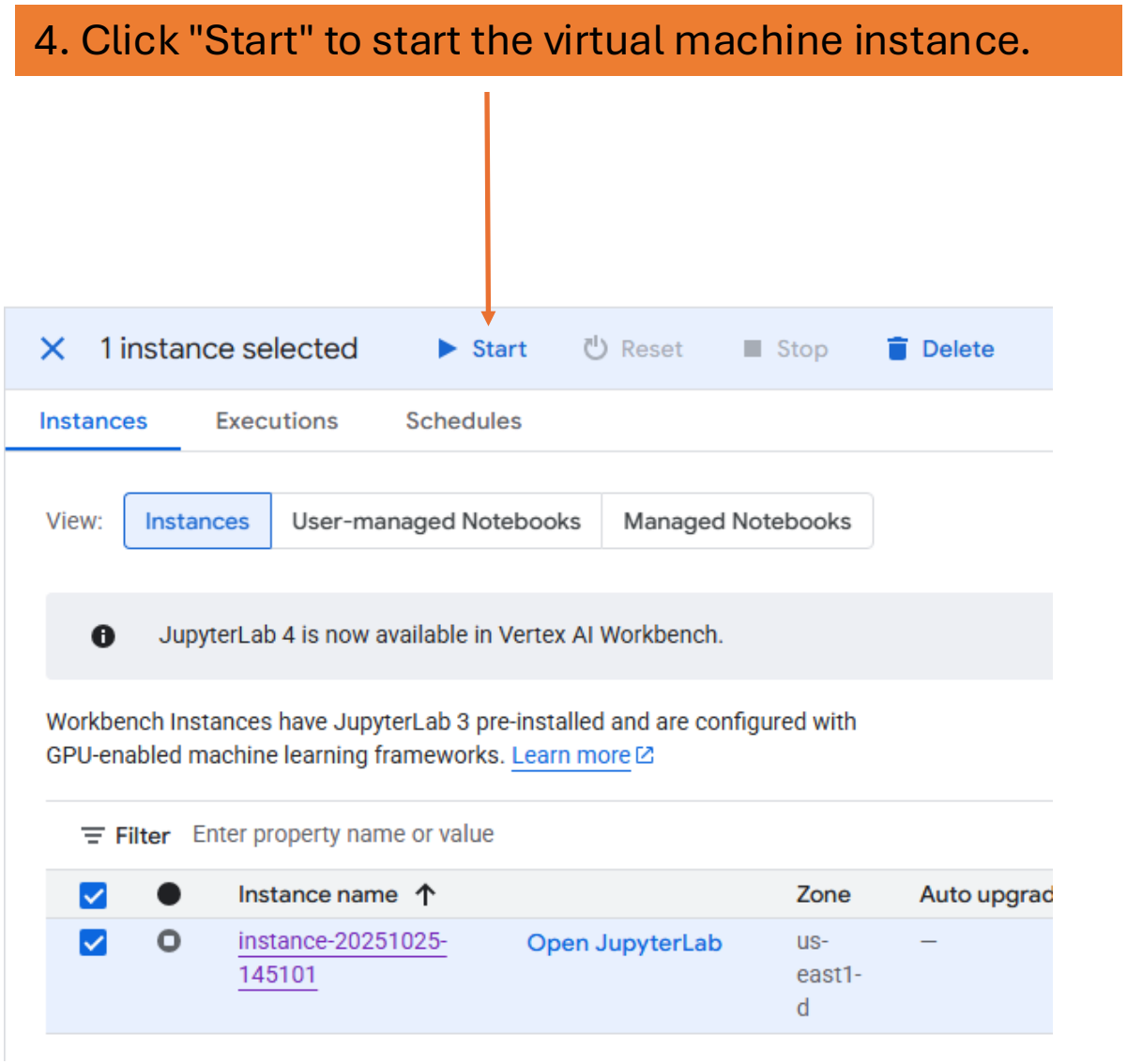


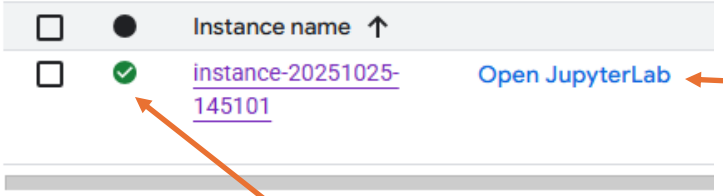
2. Navigate to your project.

1. Open the Vertex AI workbench using the link below.

https://console.cloud.google.com/vertex-ai/workbench/


Workbench
Create New
Refresh
Instances
Executions
Schedules
View:
Instances
User-managed Notebooks
Managed Notebooks
JupyterLab 4 is now available in Vertex AI Workbench.
Workbench Instances have JupyterLab 3 pre-installed and are configured with GPU-enabled machine learning frameworks. Learn more
Filter
Enter property name or value
Instance name
Zone
Auto upgrade
Version
instance-20251025-145101
Open JupyterLab
us-east1-d
M134
3. Click the checkbox.

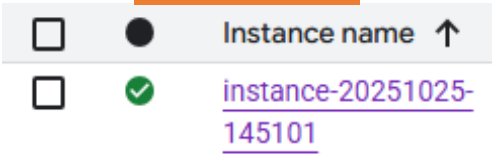

4. Click "Start" to start the virtual machine instance.
1 instance selected
Start
Reset
Stop
Delete
Instances
Executions
Schedules
View:
Instances
User-managed Notebooks
Managed Notebooks
JupyterLab 4 is now available in Vertex AI Workbench.
Workbench Instances have JupyterLab 3 pre-installed and are configured with GPU-enabled machine learning frameworks. Learn more
Filter
Enter property name or value
Instance name
Zone
instance-20251025-145101
Open JupyterLab
us-east1-d



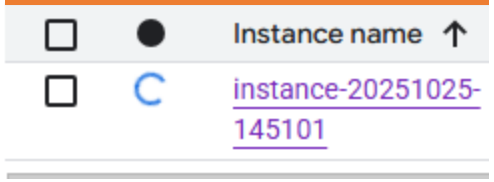
5. Once you see the green check mark, click "Open JupyterLab".

Note: You can see the instance status by hovering over this symbol. If it is a gray stop button, the VM instance is off. If there is a check mark, the VM instance is active. You will see a loading symbol here whenever powering on or powering down the instance.

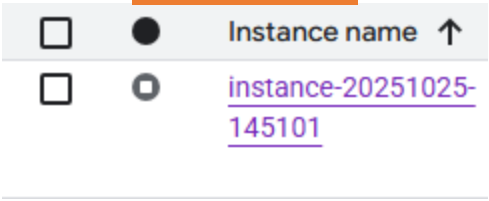
Active

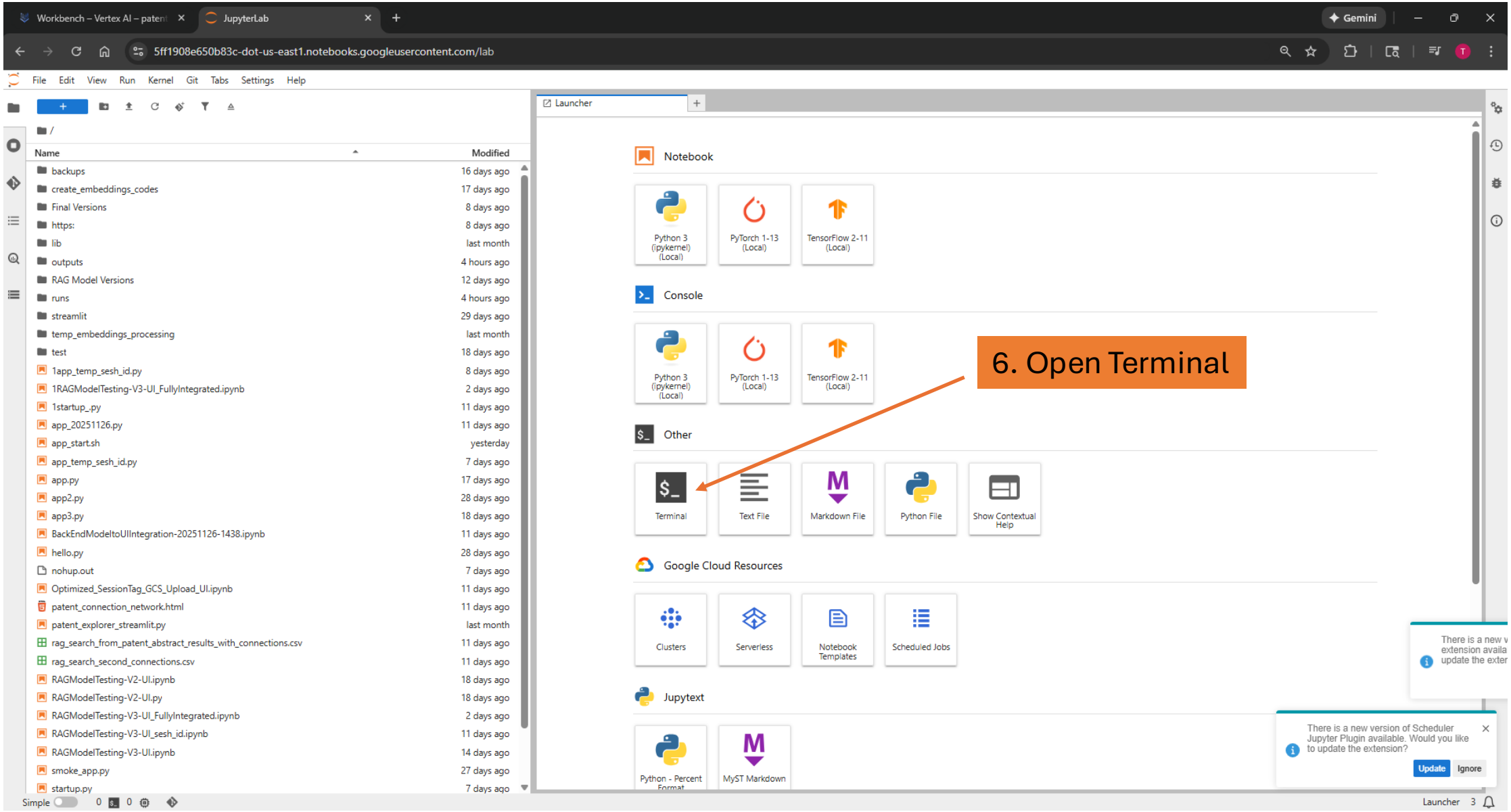


Loading/Provisioning



Stopped




Workbench – Vertex AI – patent
JupyterLab
Gemini
5ff1908e650b83c-dot-us-east1.notebooks.googleusercontent.com/lab
File Edit View Run Kernel Git Tabs Settings Help
Name
Modified
backups 16 days ago
create_embeddings_codes 17 days ago
Final Versions 8 days ago
https: 8 days ago
lib last month
outputs 4 hours ago
RAG Model Versions 12 days ago
runs 4 hours ago
streamlit 29 days ago
temp_embeddings_processing last month
test 18 days ago
1app_temp_sesh_id.py 8 days ago
1RAGModelTesting-V3-UI_FullyIntegrated.ipynb 2 days ago
1startup_.py 11 days ago
app_20251126.py 11 days ago
app_start.sh yesterday
app_temp_sesh_id.py 7 days ago
app.py 17 days ago
app2.py 28 days ago
app3.py 18 days ago
BackEndModeltoUIIntegration-20251126-1438.ipynb 11 days ago
hello.py 28 days ago
nohup.out 7 days ago
Optimized_SessionTag_GCS_Upload_UI.ipynb 11 days ago
patent_connection_network.html 11 days ago
patent_explorer_streamlit.py last month
rag_search_from_patent_abstract_results_with_connections.csv 11 days ago
rag_search_second_connections.csv 11 days ago
RAGModelTesting-V2-UI.ipynb 18 days ago
RAGModelTesting-V2-UI.py 18 days ago
RAGModelTesting-V3-UI_FullyIntegrated.ipynb 2 days ago
RAGModelTesting-V3-UI_sesh_id.ipynb 11 days ago
RAGModelTesting-V3-UI.ipynb 14 days ago
smoke_app.py 27 days ago
startup.py 7 days ago
Launcher
Notebook
Python 3 (ipykernel) (Local)
PyTorch 1-13 (Local)
TensorFlow 2-11 (Local)
Console
Python 3 (ipykernel) (Local)
PyTorch 1-13 (Local)
TensorFlow 2-11 (Local)
Other
Terminal
Text File
Markdown File
Python File
Show Contextual Help
Google Cloud Resources
Clusters
Serverless
Notebook Templates
Scheduled Jobs
Jupytext
Python - Percent Format
MyST Markdown
6. Open Terminal
There is a new version of Scheduler Jupyter Plugin available. Would you like to update the extension?
Update
Ignore
Simple
Launcher 3

Terminal 1

```
(base) jupyter@instance-20251025-145101:~$
```

7. Wait for the $ to appear.

Terminal 1

```
(base) jupyter@instance-20251025-145101:~$ ./app_start.sh
```

8. Type "./app_start.sh" and hit Enter.

Terminal 1

```
(base) jupyter@instance-20251025-145101:~$ ./app_start.sh
Streamlit ready on port 8501
Found config

Streamlit started (PID 9888). Please wait for the link...

🔗Click the link below to launch the app!
(It may take a few seconds for the link to appear.)
https://relevance-mailman-ppm-florence.trycloudflare.com
```

9. Now click on this cloudflare link.

Workbench – Vertex AI – paten | lab - JupyterLab | NASA TTO Patent Matching Too

relevance-mailman-ppm-florence.trycloudflare.com

NASA Technology Transfer Office

AI Tool for Patent Matching

Enter details on the left, click **Run Search**, and see results on the right.

Input

Document Code (CC-<digits>-<2 alphanumerics>)

EP-1234567-A1 or ep1234567a1

Search Type

auto

Results to retrieve (Top-K)

Set a number All

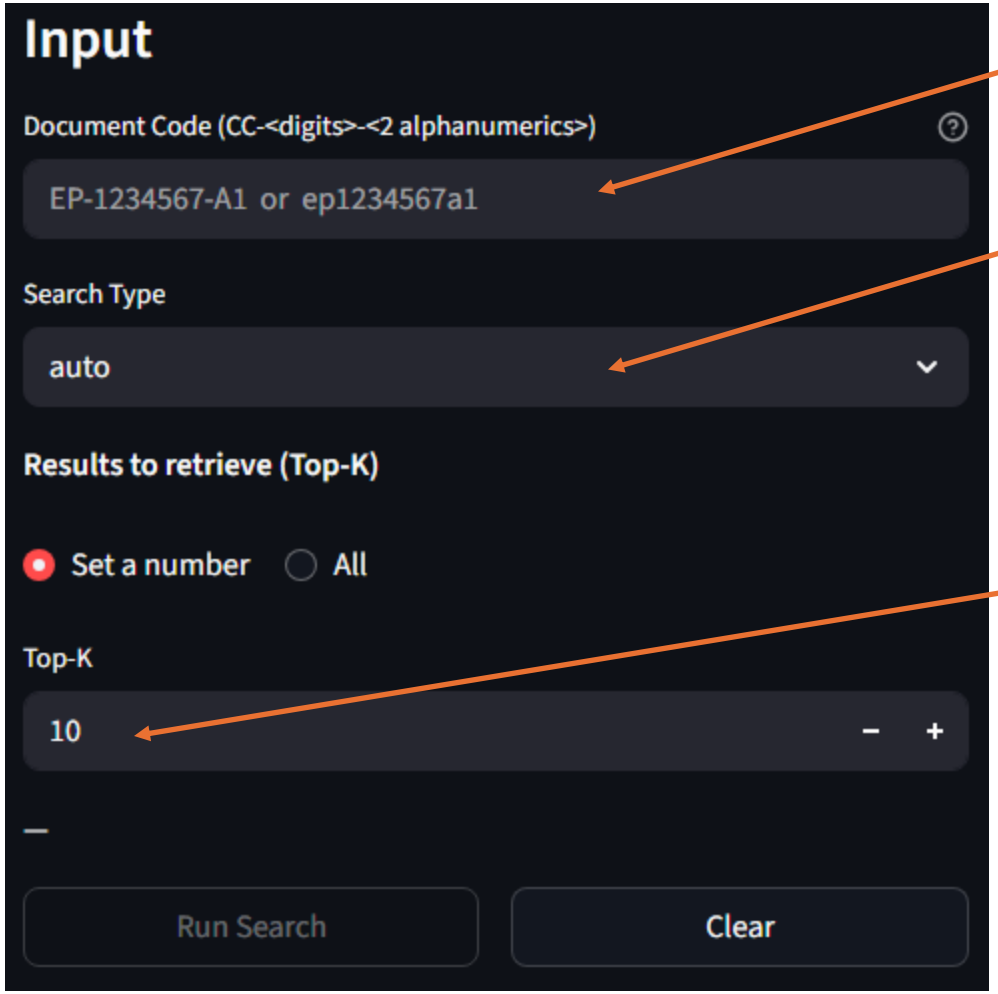
Top-K

100 – +

Output

Enter inputs and click Run Search to generate outputs.

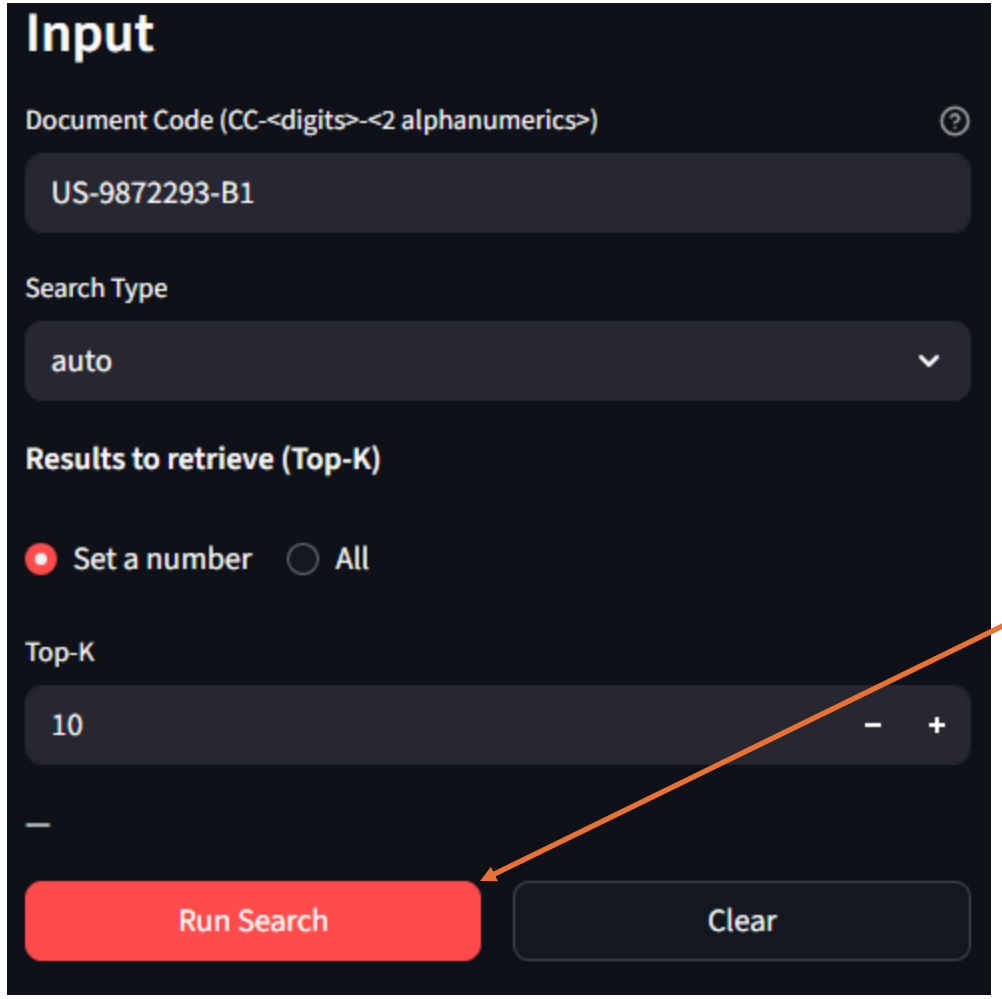
Welcome to the User Interface!



10. Enter the patent number you would like to search

11. (Optional) Use the dropdown if you would like to specify if your entered patent number is a publication/application number.

12. Top-k is the number of first degree results you will see in the output. We recommend using 10 to start off. The higher your top-k value, the longer the tool will take to produce results.



13. Click on "Run Search".

Workbench – Vertex AI – paten | lab - JupyterLab | NASA TTO Patent Matching Too

relevance-mailman-ppm-florence.trycloudflare.com

Stop

NASA Technology Transfer Office

AI Tool for Patent Matching

Enter details on the left, click **Run Search**, and see results on the right.

Note this indicator that the model is running.

Input

Document Code (CC-<digits>-<2 alphanumerics>)

US-9872293-B1

Search Type

auto

Results to retrieve (Top-K)

Set a number All

Top-K

10

Output

Running the core similarity search model.. • Elapsed 01:01

Cancel run

You can see how much time has elapsed.

You can also cancel the run.

Once the model is finished, 4 main sections will output.

1. Google Patents web scraping for patent information

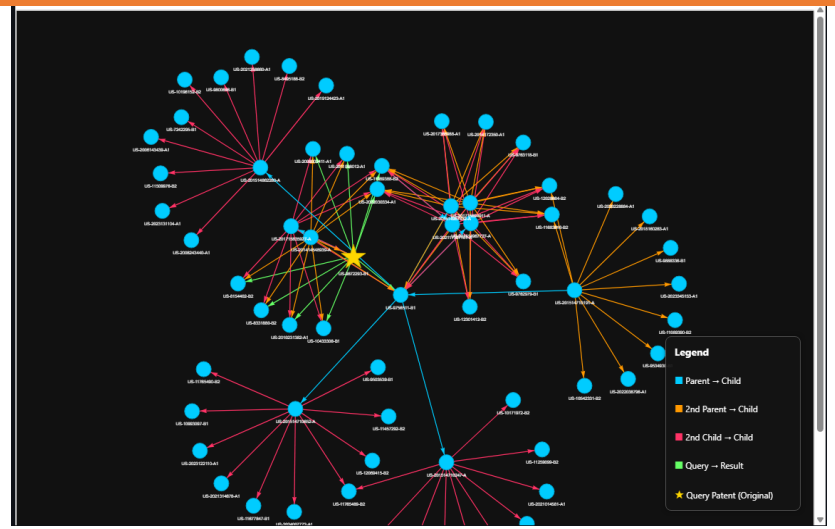
US-9872293-B1 : Intelligent data transfer for multiple sensor networks over a broad temperature range

Abstract:

A sensor network may be configured to operate in extreme temperature environments. A sensor may be configured to generate a frequency carrier, and transmit the frequency carrier to a node. The node may be configured to amplitude modulate the frequency carrier, and transmit the amplitude modulated frequency carrier to a receiver.

- **Publication:** US9872293B1
- **Application:** US14/548,939
- **Grant:** —
- **Inventor(s):** Michael Krasowski
- **Current assignee(s):** National Aeronautics and Space Administration NASA
- **Filed:** 2014-11-20
- **Published:** 2014-11-20
- **Granted:** granted
- **Google Patents:** https://patents.google.com/patent/US9872293B1/en

2. An Interactive Patent Connection Network



3. CSV files showing all patent results

< 0.6 | 0.6–0.8 | 0.8–0.9 | > 0.9

	Rank	Match	Similarity_Score	Publication_Number	Application_Number	Title	Inventors	A
0	1		0.8423	US-9872293-B1	US-201414548939-A	Intelligent data transfer for multiple sensor networks over a broad temperature range	[{'name': 'KRASOWSKI MICHAEL', 'country_code': 'US'}]	[{
1	2		0.8423	US-10433308-B1	US-201715835927-A	Intelligent data transfer for multiple sensor networks over a broad temperature range	[{'name': 'KRASOWSKI MICHAEL J', 'country_code': 'US'}]	[{
2	3		0.6872	US-2009003411-A1	US-13282208-A	High-temperature sensing system with passive wireless communication	[{'name': 'SCHUDA FELIX J', 'country_code': 'US'}]	[{
3	4		0.6609	US-8154402-B2	US-40331809-A	Wireless temperature sensor network	[{'name': 'TAYRANI REZA', 'country_code': 'US'} {'name': 'YOUNG RICHARD D', 'country	[{
4	5		0.6609	US-2010231382-A1	US-40331809-A	Wireless temperature sensor network	[{'name': 'TAYRANI REZA', 'country_code': 'US'} {'name': 'YOUNG RICHARD D', 'country	[{
5	6		0.6543	US-8331880-B2	US-75936510-A	Transmitter with self-test capability	[{'name': 'SORNIN NICOLAS', 'country_code': 'FR'}]	[{
6	7		0.6543	US-2011098012-A1	US-75936510-A	Transmitter with Self-Test Capability	[{'name': 'SORNIN NICOLAS', 'country_code': 'FR'}]	[{
7	8		0.6513	US-9756511-B1	US-201514710209-A	System, method and apparatus for wireless sensor network configuration	[{'name': 'STAMATAKIS JULIEN G', 'country_code': 'US'} {'name': 'AL-MOMEN SERENE'	[{
8	9		0.6513	US-2022030334-A1	US-202117397161-A	System, Method and Apparatus for Wireless Sensor Network Configuration	[{'name': 'STAMATAKIS JULIEN G', 'country_code': 'US'} {'name': 'AL-MOMEN SERENE'	[{
9	10		0.6513	US-11089388-B2	US-201715694911-A	System, method and apparatus for wireless sensor network configuration	[{'name': 'STAMATAKIS JULIEN G', 'country_code': 'US'} {'name': 'AL-MOMEN SERENE'	[{

4. Statistical plots for the results

Input

Document Code (CC-<digits>-<2 alphanumerics>)

US-9872293-B1

Search Type

auto

Results to retrieve (Top-K)

Set a number All

Top-K

10

—

Run Search Clear

How these results are generated:

When you click **Run Search**, the app looks up the patent you entered, uses its text (mainly the abstract) to understand what it's about, then finds other patents with the most similar language and topics. The goal is to quickly surface related prior art and nearby ideas—not to

Output

Total elapsed ti

You can download these outputs in a zipped file for later use.

Download all outputs (.zip)

US-9872293-B1 : Intelligent data transfer for multiple sensor networks over a broad temperature range

Abstract:

A sensor network may be configured to operate in extreme temperature environments. A sensor may be configured to generate a frequency carrier, and transmit the frequency carrier to a node. The node may be configured to amplitude modulate the frequency carrier, and transmit the amplitude modulated frequency carrier to a receiver.

- **Publication:** US9872293B1
- **Application:** US14/548,939
- **Grant:** —
- **Inventor(s):** Michael Krasowski
- **Current assignee(s):** National Aeronautics and Space Administration NASA
- **Filed:** 2014-11-20
- **Published:** 2014-11-20
- **Granted:** granted
- **Google Patents:** https://patents.google.com/patent/US9872293B1/en

Interactive Graph

For the CSV files, the primary results will show query to result only. (1st connection)

Primary Results

< 0.6 | 0.6–0.8 | 0.8–0.9 | > 0.9

	Rank	Match	Similarity_Score	Publication_Number	Application_Number	Title	Inventors
0	1		0.8423	US-9872293-B1	US-201414548939-A	Intelligent data transfer for multiple sensor networks over a broad temperature range	[{'name': 'KRASOWSKI MICHAEL', 'country_code': 'US'}]
1	2		0.8423	US-10433308-B1	US-201715835927-A	Intelligent data transfer for multiple sensor networks over a broad temperature range	[{'name': 'KRASOWSKI MICHAEL J', 'country_code': 'US'}]
2	3		0.6872	US-2009003411-A1	US-13282208-A	High-temperature sensing system with passive wireless communication	[{'name': 'SCHUDA FELIX J', 'country_code': 'US'}]
3	4		0.6609	US-8154402-B2	US-40331809-A	Wireless temperature sensor network	[{'name': 'TAYRANI REZA', 'country_code': 'US'} {'name': 'YOUNG RICHAI
4	5		0.6609	US-2010231382-A1	US-40331809-A	Wireless temperature sensor network	[{'name': 'TAYRANI REZA', 'country_code': 'US'} {'name': 'YOUNG RICHAI
5	6		0.6543	US-8331880-B2	US-75936510-A	Transmitter with self-test capability	[{'name': 'SORNIN NICOLAS', 'country_code': 'FR'}]
6	7		0.6543	US-2011098012-A1	US-75936510-A	Transmitter with Self-Test Capability	[{'name': 'SORNIN NICOLAS', 'country_code': 'FR'}]
7	8		0.6513	US-9756511-B1	US-201514710209-A	System, method and apparatus for wireless sensor network configuration	[{'name': 'STAMATAKIS JULIEN G', 'country_code': 'US'} {'name': 'AL-MO
8	9		0.6513	US-2022030334-A1	US-202117397161-A	System, Method and Apparatus for Wireless Sensor Network Configuration	[{'name': 'STAMATAKIS JULIEN G', 'country_code': 'US'} {'name': 'AL-MO
9	10		0.6513	US-11089388-B2	US-201715694911-A	System, method and apparatus for wireless sensor network configuration	[{'name': 'STAMATAKIS JULIEN G', 'country_code': 'US'} {'name': 'AL-MO

Download original CSV

Second-Level Results

< 0.6 | 0.6–0.8 | 0.8–0.9 | > 0.9

	Rank	Match	Similarity_Score	Publication_Number	Application_Number	Title	Inventors
7	7		0.8279	US-12028664-B2	US-202318208982-A	Method and apparatus for remote configuration of sensor data collection	[{'name': 'STAMATAKIS JULIEN G', 'country_code': 'US'}]
8	8		0.8057	US-11683616-B2	US-202117396953-A	System, method and apparatus for remote wireless sensor device configuration	[{'name': 'STAMATAKIS JULIEN G', 'country_code': 'US'}]
9	9		0.7352	US-12301412-B2	US-202318228096-A	Method and apparatus for monitoring a configuration status of a remote sensor devic	[{'name': 'STAMATAKIS JULIEN G', 'country_code': 'US'} {'name': 'HOFFI
10	10		0.7139	US-2014172350-A1	US-201314108698-A	Remote sensor configuration	[{'name': 'LOTHAMER JEFFREY P', 'country_code': 'US'}]
11	1		0.9368	US-11089388-B2	US-201715694911-A	System, method and apparatus for wireless sensor network configuration	[{'name': 'STAMATAKIS JULIEN G', 'country_code': 'US'} {'name': 'AL-MO
12	2		0.9368	US-9756511-B1	US-201514710209-A	System, method and apparatus for wireless sensor network configuration	[{'name': 'STAMATAKIS JULIEN G', 'country_code': 'US'} {'name': 'AL-MO
13	3		0.9368	US-2022030334-A1	US-202117397161-A	System, Method and Apparatus for Wireless Sensor Network Configuration	[{'name': 'STAMATAKIS JULIEN G', 'country_code': 'US'} {'name': 'AL-MO
14	4		0.9368	US-2017366988-A1	US-201715694911-A	System, Method and Apparatus for Wireless Sensor Network Configuration	[{'name': 'STAMATAKIS JULIEN G', 'country_code': 'US'} {'name': 'AL-MO
15	5		0.9055	US-9762979-B1	US-201514967727-A	System, method and apparatus for operating mode configuration in a wireless sensor	[{'name': 'STAMATAKIS JULIEN G', 'country_code': 'US'} {'name': 'AL-MO
16	6		0.8916	US-9763118-B1	US-201514967832-A	System, method and apparatus for metering configuration in a wireless sensor netwc	[{'name': 'STAMATAKIS JULIEN G', 'country_code': 'US'} {'name': 'AL-MO
17	7		0.8279	US-12028664-B2	US-202318208982-A	Method and apparatus for remote configuration of sensor data collection	[{'name': 'STAMATAKIS JULIEN G', 'country_code': 'US'}]

Note: Make sure to use the horizontal scroll bar. The CSV outputs will have the following columns:

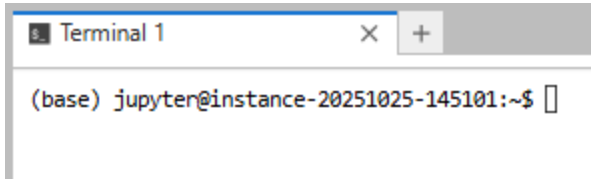
- Rank
- Match
- Similarity Score
- Publication/Application Number
- Title
- Inventors
- Assignees
- Parent/Child Connections
- CPC Codes
- Top Keywords
- Abstract

The secondary results will show 2nd and 3rd degree connections.

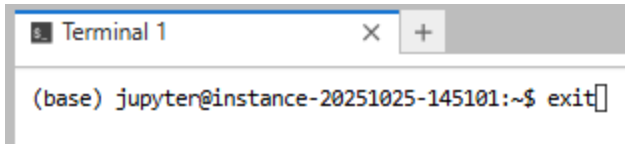
The Level 1 visual plots will show query to result only. (1st connection)

The Level 2 visual plots will show 2nd and 3rd degree connections.

Once you are finished and ready to shut down the VM instance...

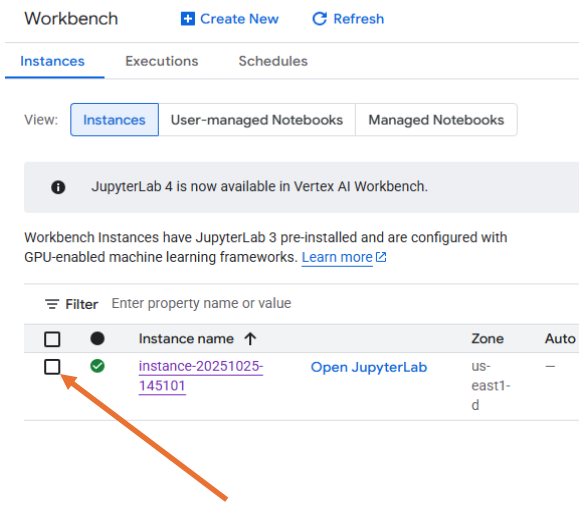


1. Go back to the terminal you were using.

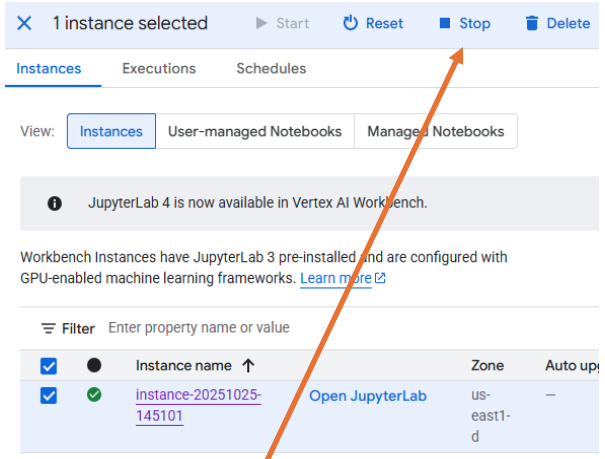


2. Type "exit" and hit Enter. This will shut down the terminal.

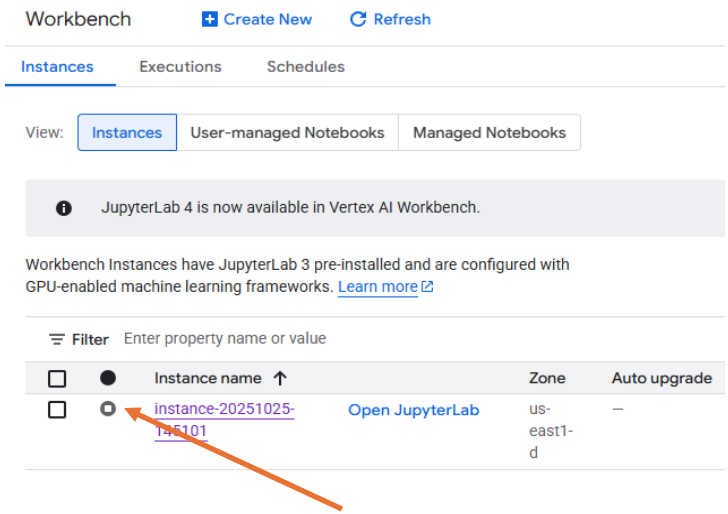
3. Close your JupyterLab tab in your browser.



4. Click the checkbox.



5. Click Stop.



The VM is now turned off.